

The Financial Seismograph: Decoding Systemic Risk with Linear Algebra

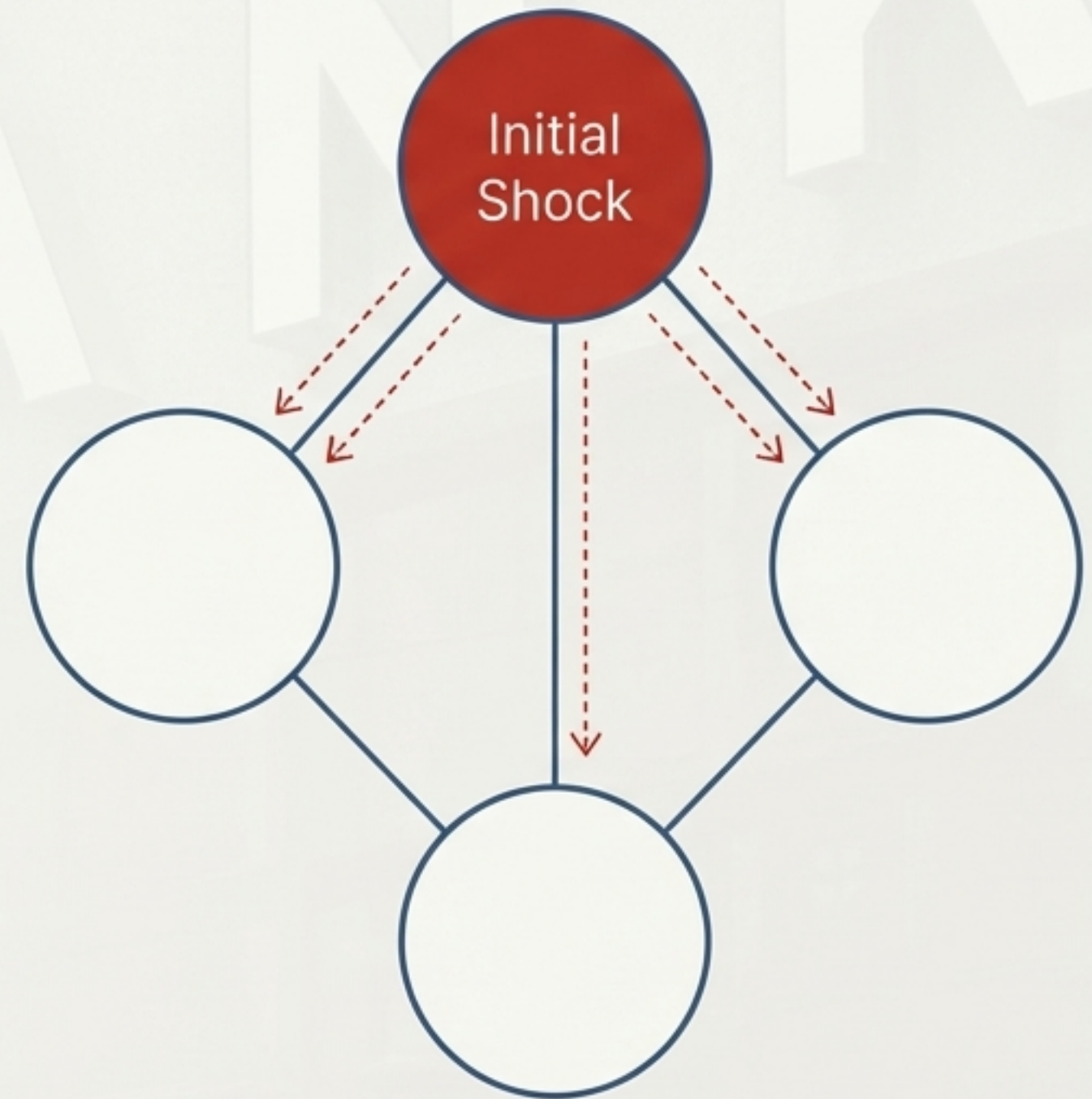
An analysis of how hidden mathematical structures reveal the stability of interconnected financial networks.

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When One Bank Falters, The System Trembles

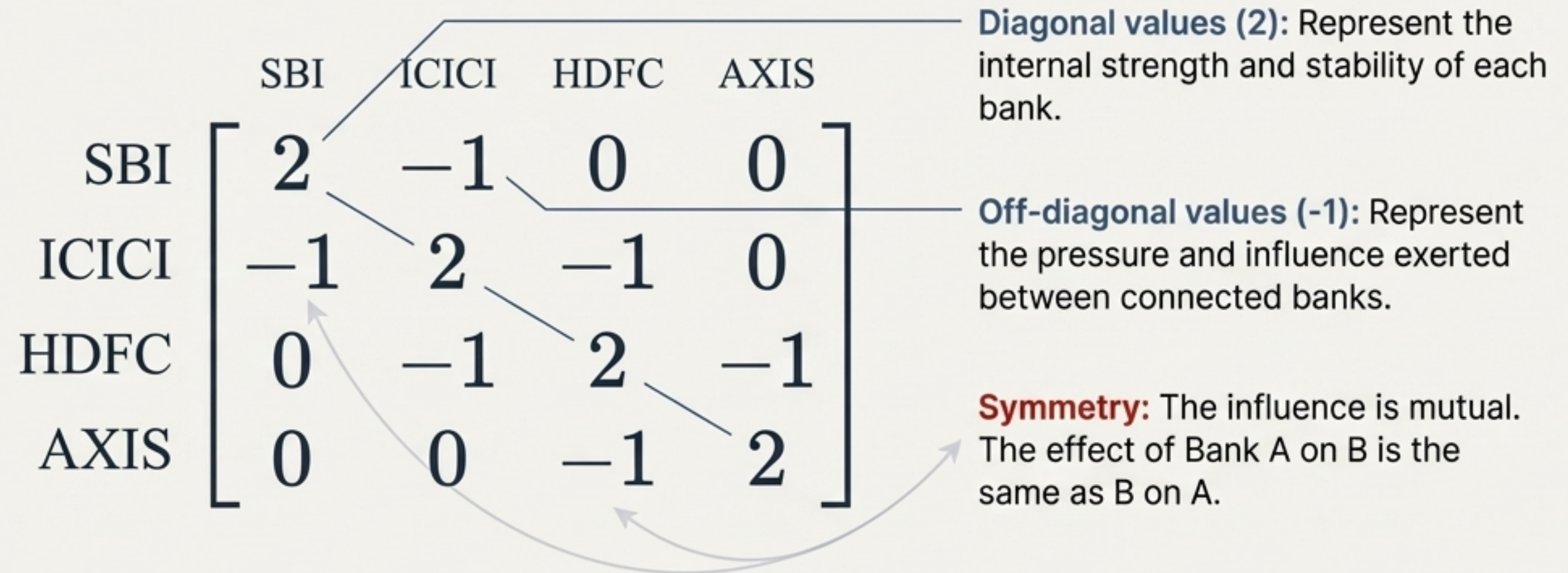
Banks do not operate in isolation. They are intricately linked through loans, investments, and market activities. This interconnectedness means that a shock to one institution—or the market as a whole—doesn't stay contained. It propagates through the network, creating a domino effect known as **systemic risk**.

How can we understand and measure the hidden pathways through which this financial contagion spreads?



Mapping the Financial Network with a Matrix

We can model the banking system as a mathematical object: a matrix. Each row and column represents a specific bank (e.g., SBI, ICICI, HDFC, Axis). The values within the matrix define the relationships between them.



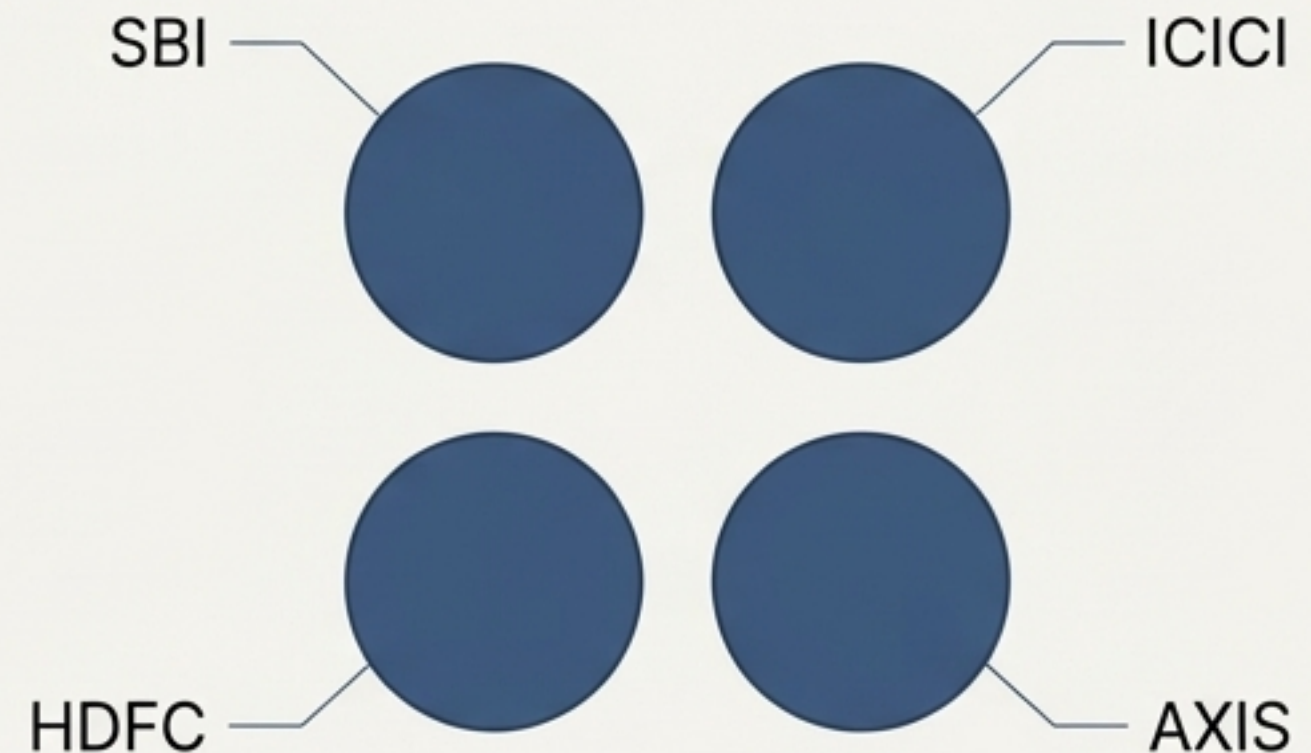
The System's DNA: What the Matrix Reveals

Reading 1: Rank — A System of Individuals

Key Finding: The Rank of Matrix A is 4.

Financial Interpretation: A rank of 4 in a 4-bank system signifies that each bank is independent. No institution's behavior is a mere shadow or combination of the others. Every bank plays a unique and non-redundant role in the financial network.

```
#RANK
rank_A = np.linalg.matrix_rank(A)
print("Rank of Matrix A is :",rank_A,"\n")
... Rank of Matrix A is : 4
```



The System's DNA: Stability and Strength

Reading 2: Determinant — A Stable System

Key Finding: The Determinant of Matrix A is 5.0.

Financial Interpretation: A non-zero determinant means the system is stable and invertible. For any given market shock, there is a single, unique, and calculable outcome for the banks. The system won't collapse into chaos or ambiguity.

```
#Determinant
det_A = np.linalg.det(A).round()
print("Determinant of Matrix A is : ",det_A,"\n")
... Determinant of Matrix A is : 5.0
```

Reading 3: Trace — Balanced Internal Strength

Key Finding: The Trace of A is 8.

Financial Interpretation: The trace represents the sum of the system's internal strengths (the diagonal elements). A value of 8 indicates a balanced and significant total internal stability across all four banks.

```
#Trace
trace_A = np.trace(A)
print("Trace of A is : ",trace_A,"\n")
... Trace of A is : 8
```

The System's DNA: No Shock Goes Unfelt

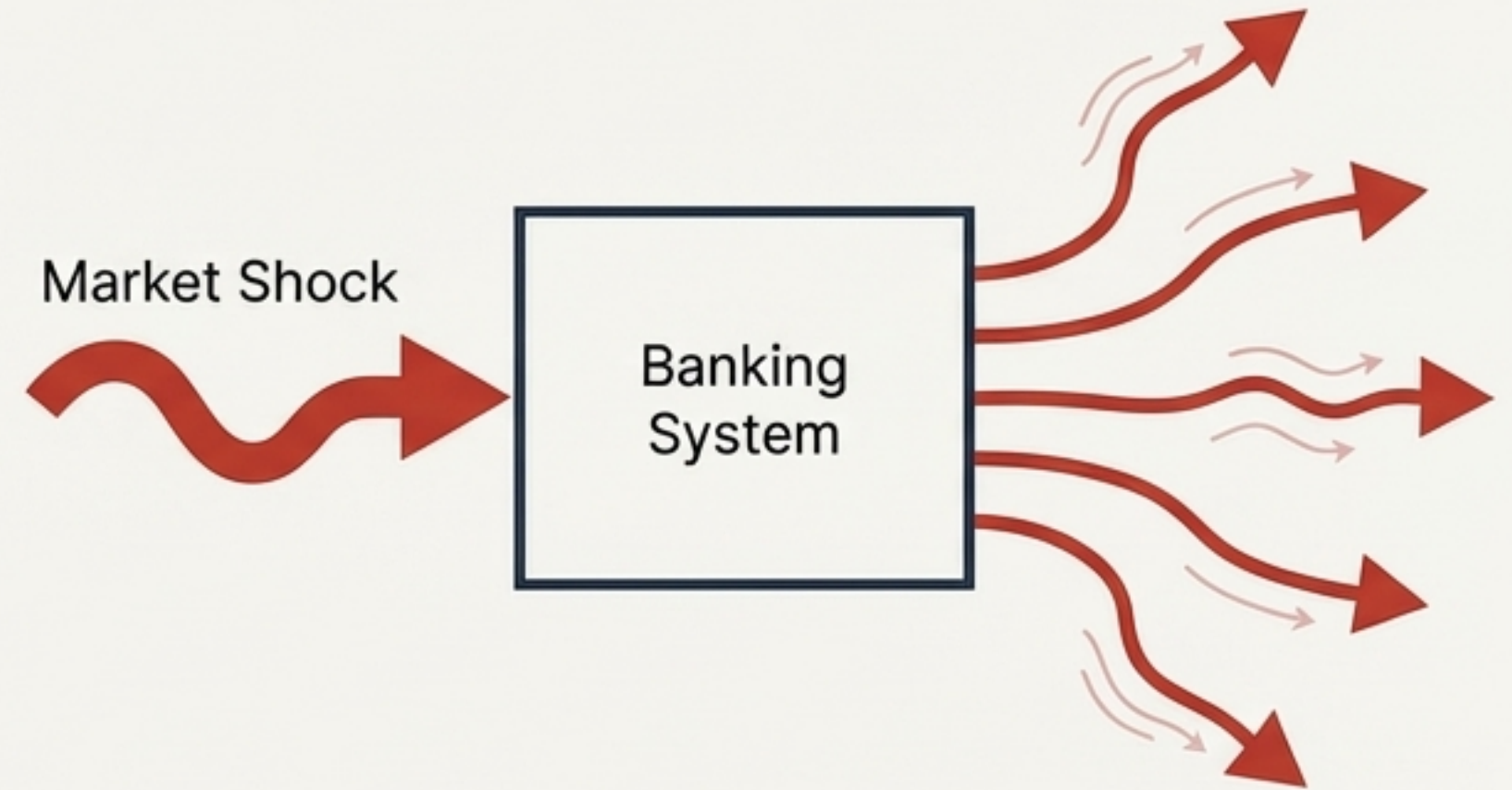
Reading 4: Null Space — A Highly Sensitive Network

Key Finding: The Null Space of Matrix A is empty.

Financial Interpretation: This is a critical insight. An empty null space confirms that the system is fully responsive. There is no type of **external market shock** that can be introduced and simply "disappear" or be absorbed without causing a reaction. Every market event, no matter its form, **will create a ripple effect**.

```
#Null Space
null_space_A = null_space(A)
print("Null Space of A is : \n",null_space_A,"\n")

... Null Space of A is :
[]
```



Simulating a Market Tremor: How Banks Respond

We introduce an external market shock, represented by a vector **b**. By solving the equation **Ax = b**, we can determine the resulting stress (**x**) experienced by each bank.

$$\begin{bmatrix} 2.5 & -0.5 & -0.5 & 0.0 \\ -0.5 & 2.5 & -0.5 & -0.5 \\ -0.5 & -0.5 & 2.5 & -0.5 \\ 0.0 & -0.5 & -0.5 & 2.5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 1 \end{bmatrix}$$

External Market Shock

Resulting Bank Stress (**x**):

$$\begin{bmatrix} 3 \\ 5 \\ 5 \\ 3 \end{bmatrix}$$

```
Solve_Ab = np.linalg.solve(A, vector_b)
print("Effect of external market shocks on Bank is :", Solve_Ab)
... Effect of external market shocks on Bank is : [3. 5. 5. 3.]
```

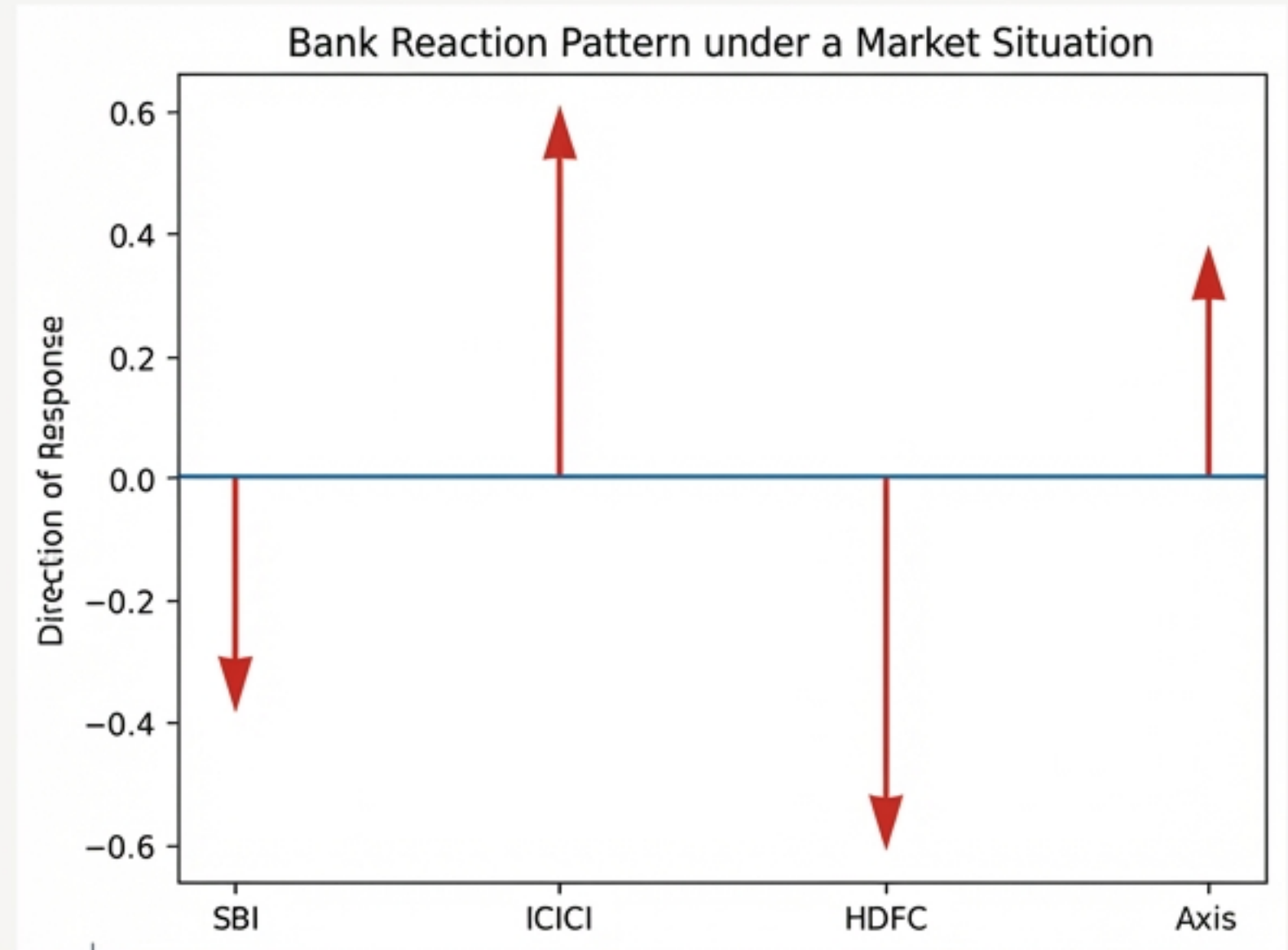
Interpretation: The shock is not felt equally. Banks in the middle of the network (ICICI, HDFC) absorb significantly more stress (5 units) than those at the edges (SBI, Axis), who feel a stress of 3 units. A bank's position in the network is as important as its individual strength.

Identifying the System's Fault Lines

Every financial network has inherent patterns of stress it is most vulnerable to. These are revealed by the matrix's eigenvalues.

Key Findings

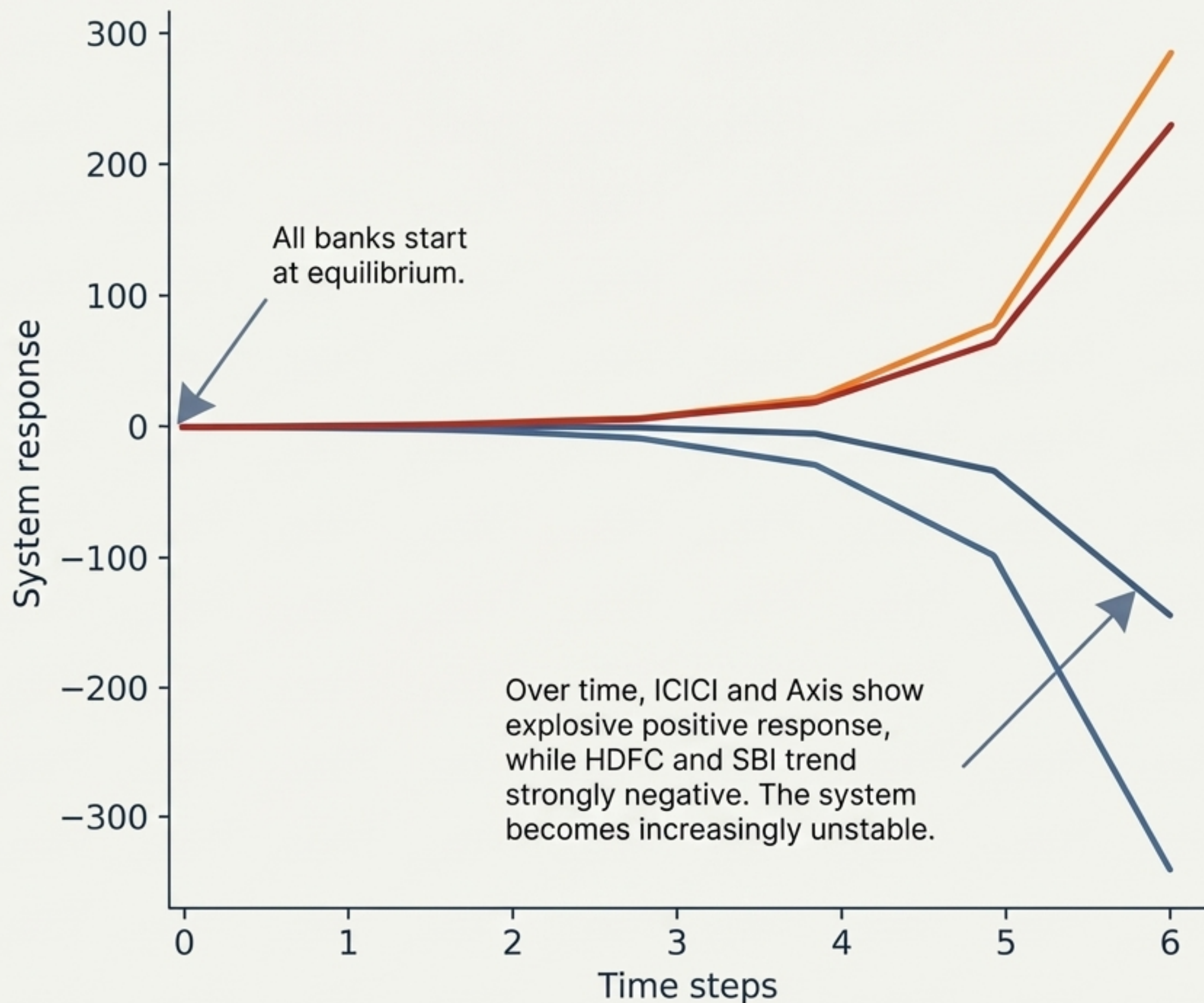
- Highest System Stress (Worst Case):
- Corresponds to the largest eigenvalue (3.618). This market situation creates maximum stress in the banking network.
- Lowest System Stress (Best Case):
- Corresponds to the smallest eigenvalue (0.382). This situation causes only minor, manageable stress.



Under a specific stress scenario, ICICI and Axis move together, while SBI and HDFC react in the opposite direction. Understanding these patterns is key to managing risk.

How a Single Shock Spreads and Amplifies Over Time

A static analysis is not enough. We simulated how an initial market stress propagates through the network over time. The results are stark: the banks' fortunes diverge dramatically. More connected banks experience accelerating pressure.



Validating the Model's Mathematical Soundness

To ensure our “seismograph” is accurately calibrated, we verified it against the Cayley-Hamilton Theorem, a fundamental principle of linear algebra. The theorem states that every matrix satisfies its own characteristic equation.

The theorem was tested in Python. The result was a matrix of values extremely close to zero, which computationally confirms the theorem holds true.

This verification confirms that our matrix-based banking model is mathematically robust and can be used to study how market stress affects connected banks.

```
[[-8.88178428e-16  7.10542736e-15 -5.59064262e-09  0.00000000e+00]
 [ 7.10542736e-15 -5.59063640e-09  7.10542736e-15 -5.59064262e-09]
 [-5.59064262e-09  7.10542736e-15 -5.59063640e-09  7.10542736e-15]
 [ 0.00000000e+00 -5.59064262e-09  7.10542736e-15 -8.88178428e-16]]
```

A Model for Understanding, Not Prediction

This project provides a powerful lens for understanding systemic behavior, but its limitations must be acknowledged.

Key Assumptions

- Simplified Model: Connections are fixed and symmetric.
- Illustrative Shocks: External market shocks are represented in a simplified vector form.

Core Purpose

The goal is not to predict exact stock prices or market behavior. The purpose of the model is to study the patterns and dynamics of how stress propagates through an interconnected financial system. Despite its simplicity, the model provides vital insights.

From Analysis to Action: Stabilizing the Network

To manage risk during economic turbulence, focus should be on the system as a whole, not just individual banks.



Monitor Key Nodes

Banks that are more central and highly connected (like ICICI and HDFC in our model) should be monitored more closely, as stress spreads fastest through them.



Strengthen Core Banks

Increase risk buffers and capital requirements for these highly connected institutions to absorb shocks before they propagate.



Implement Early Warning Systems

Use system-wide stress simulations, like the one in this project, to project, to regularly test the network's resilience to different scenarios and identify emerging vulnerabilities.

Stability is a Property of the Network, Not Just the Banks

This project demonstrated that market problems affect banks differently based on their position within the financial network. Seemingly stable individual institutions can be vulnerable due to the connections they hold.

True financial stability depends on understanding the complete system—the strength of the banks and, just as importantly, the structure of the connections between them. Linear algebra provides a powerful and elegant tool for gaining this essential system-level perspective.